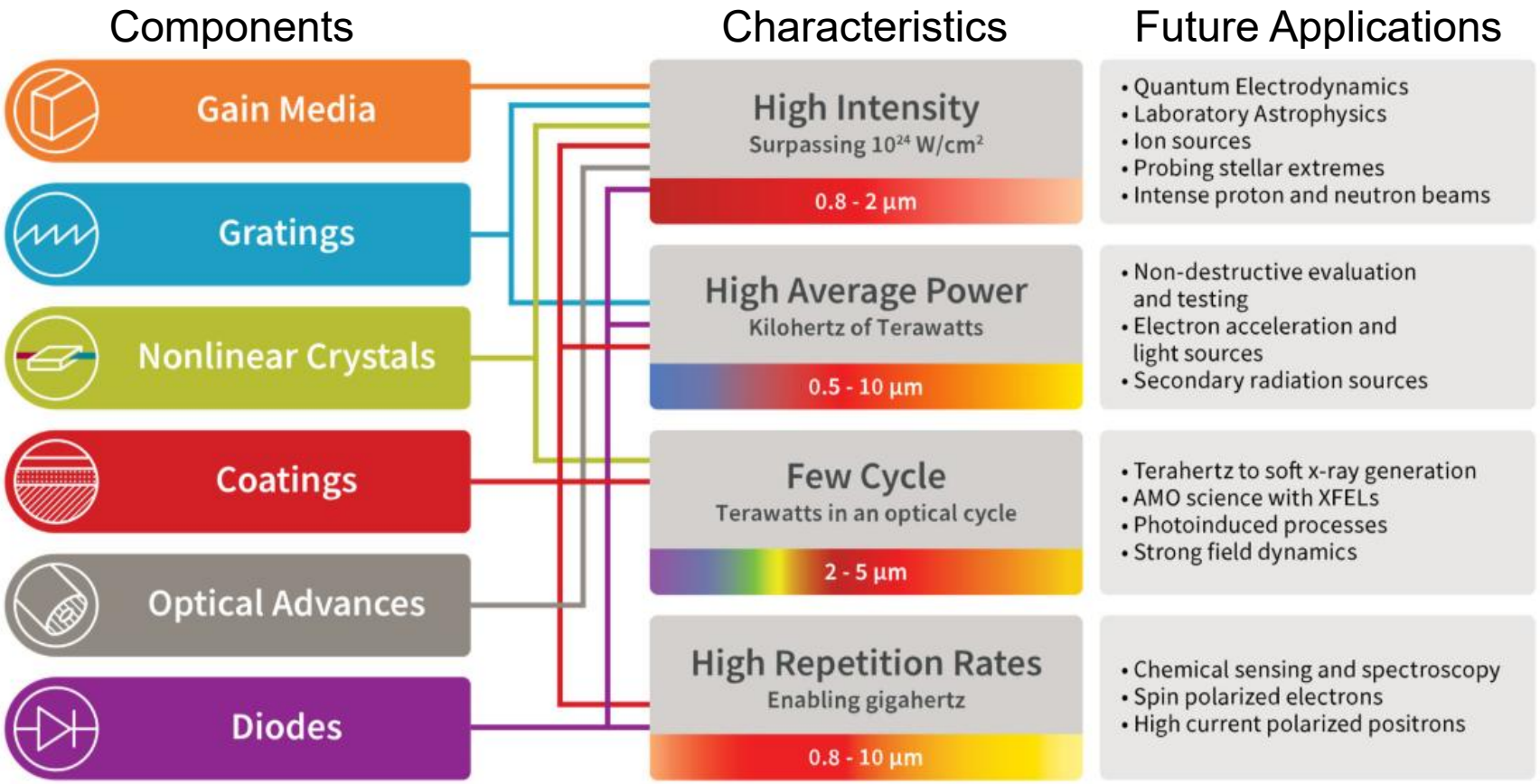
Electron Dynamics of Femtosecond Laser Induced Plasma inside Multilayer Dielectric High Reflectors Studied Using Particle-in-cell (PIC) Approach

Ziyao Su[1], Joseph R. Smith[2], Enam Chowdhury [1,3,4]

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[2] *Marietta College, Department of Physics, Marietta, OH 45750*
[3] *The Ohio State University, Department of Electrical and Computer Engineering, Columbus, OH 43210*
[4] *The Ohio State University, Department of Physics, Columbus, OH 43210*

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Motivation



Fundamental technological advances and their enabling technologies [1]

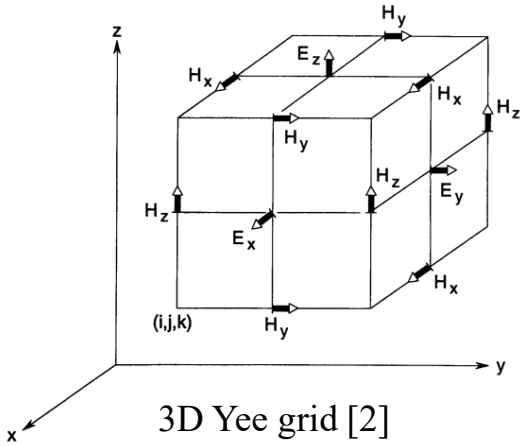
Gratings and Mirrors are identified as the weakest parts in the CPA system. Therefore, it is crucial to increase the Laser Induced Damage Threshold (LIDT) of these components.

[1] Matthias F. Kling, REPORT OF THE BASIC RESEARCH NEEDS WORKSHOP ON LASER TECHNOLOGY, 2023

Method

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FDTD



3D Yee grid [2]

Maxwell equations

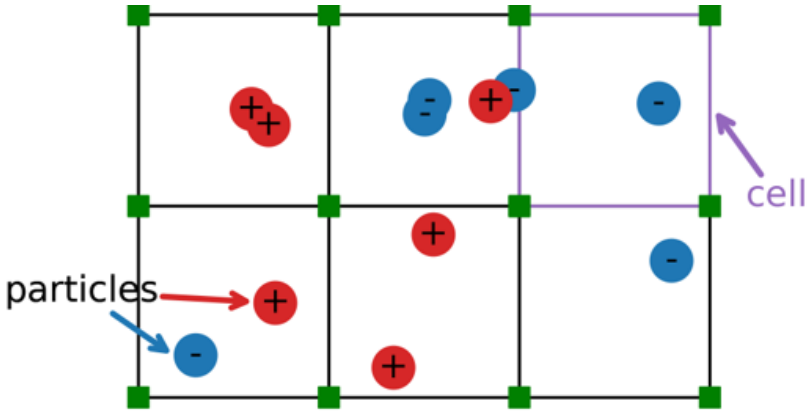
$$\nabla \times E = \frac{-\partial B}{\partial t}$$

$$\nabla \times H = \frac{\partial D}{\partial t} + J$$

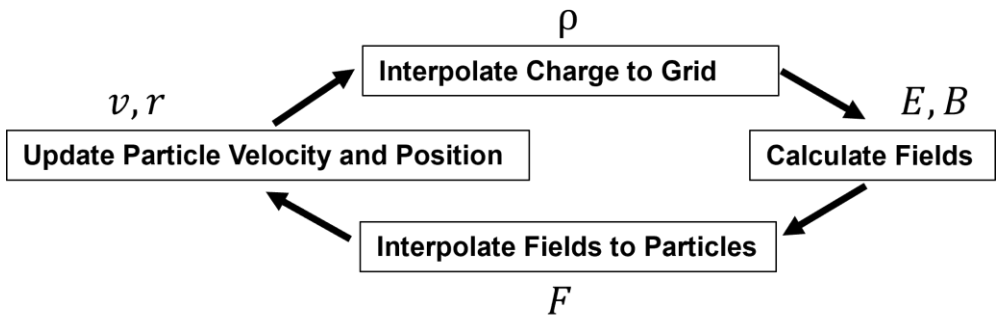
$$D = \epsilon E$$

$$B = \mu H$$

PIC



Sketch of a 2D PIC simulation grid



Both Finite Difference Time Domain (FDTD) and PIC are based on mesh grid approaches for solving electromagnetic fields, but FDTD cannot reveal the particle dynamics, while Particle In Cell (PIC) can.

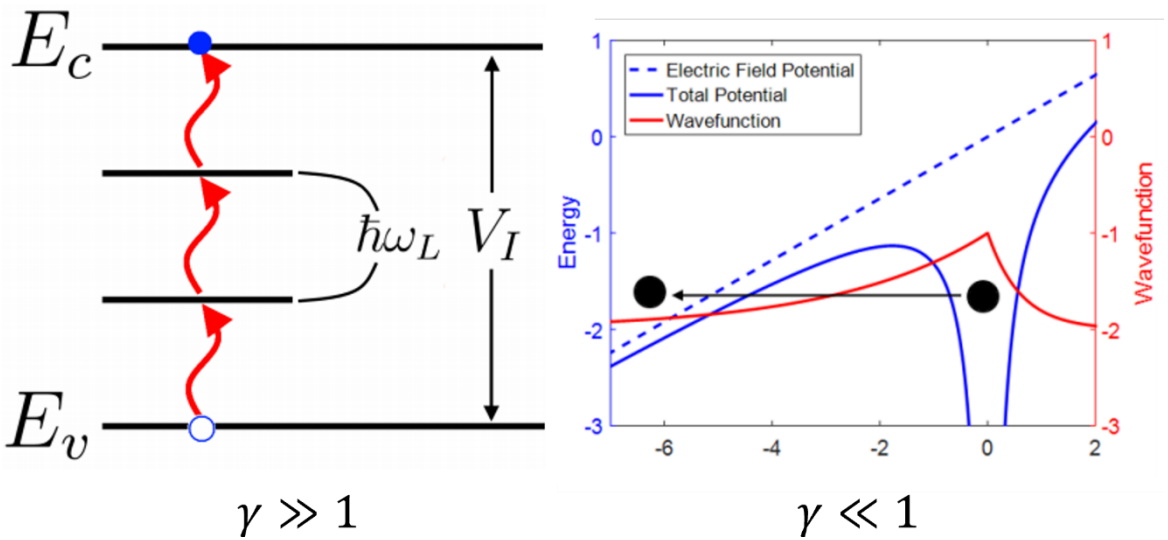
[2] Kane Yee, "Numerical solution of initial boundary value problems involving maxwell's equations in isotropic media," in *IEEE Transactions on Antennas and Propagation*, vol. 14, no. 3, pp. 302-307, May 1966, doi: 10.1109/TAP.1966.1138693.

Keldysh Parameter [3]

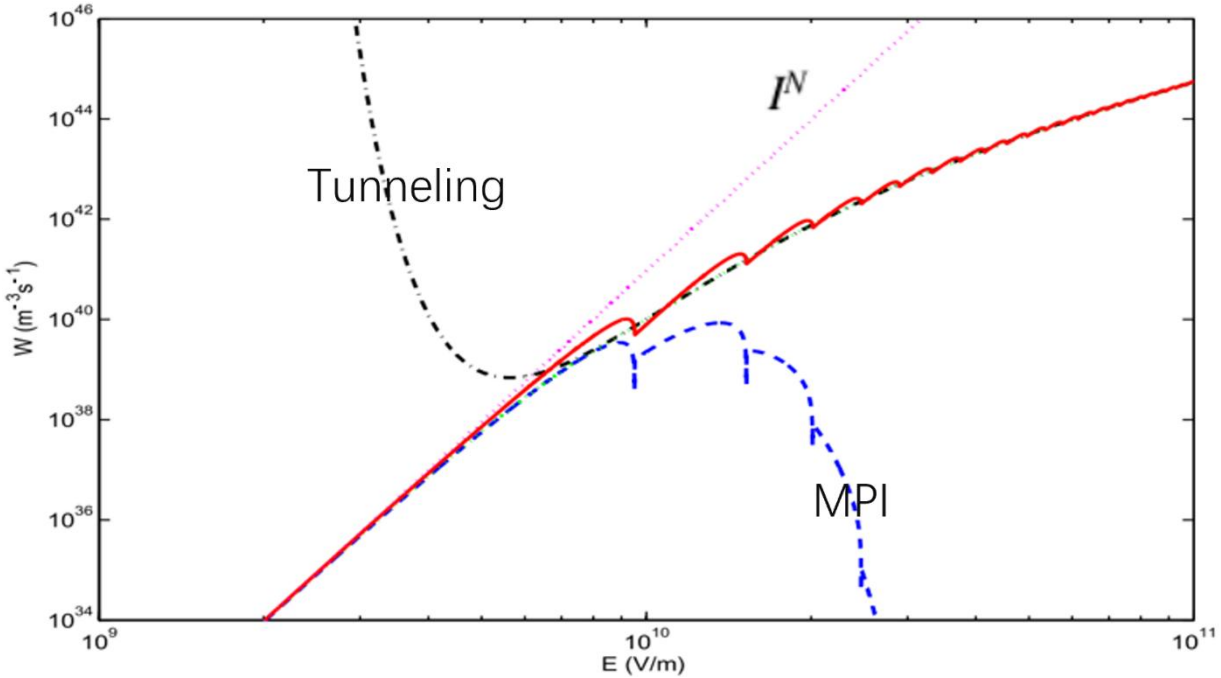
$$\gamma = \frac{\omega\sqrt{m_e W_0}}{eE}$$

W_0: Ionization energy
E: External electric field
ω : Laser frequency

Multi-photon/Tunneling Ionization Cartoon [4]



$\gamma \gg 1$ $\gamma \ll 1$



Strong Field Ionization Rates Based on Different Models (wavelength 800 nm, bandgap 6.5 eV) [5]

[3] L. V. Keldysh, "Ionization in the field of a strong electromagnetic wave," J. Exp. Theor.Phys., vol. 20, no. 5, pp. 1307–1314, 1965
[4] A. RUSSELL, "Modeling ultrashort pulse laser damage with the particle in cell method," Ph.D. dissertation, 2019.
[5] P. Balling and J. Schou, "Femtosecond-laser ablation dynamics of dielectrics: Basics and applications for thin films," Reports on Progress in Physics, vol. 76, no. 3, p. 036 502, Feb.2013. DOI: 10.1088/0034-4885/76/3/036502.

Laser-Plasma Interactions [6]



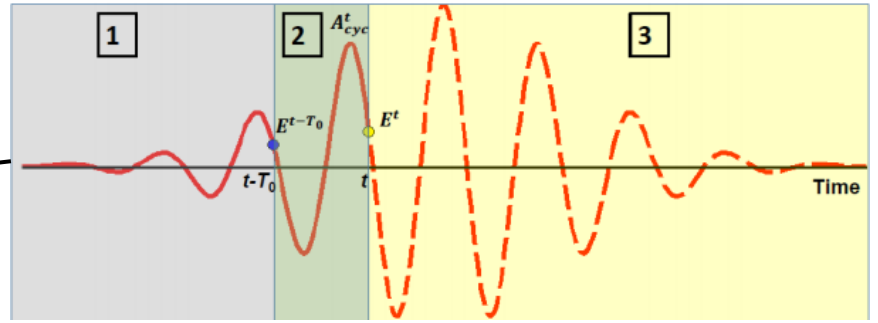
github.com/Warwick-Plasma/epoch

Optical Material Properties

Keldysh Photoionization

Impact Ionization

Cycle-averaged approach



A sketch of time profile of electric field at a fixed space point.[7]

[6] T D Arber, K Bennett, C S Brady, A Lawrence-Douglas, M G Ramsay, N J Sircombe, P Gillies, R G Evans, H Schmitz, A R Bell, "Contemporary particle-in-cell approach to laser-plasma modelling," Plasma Physics and Controlled Fusion, 2015.
[7] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)

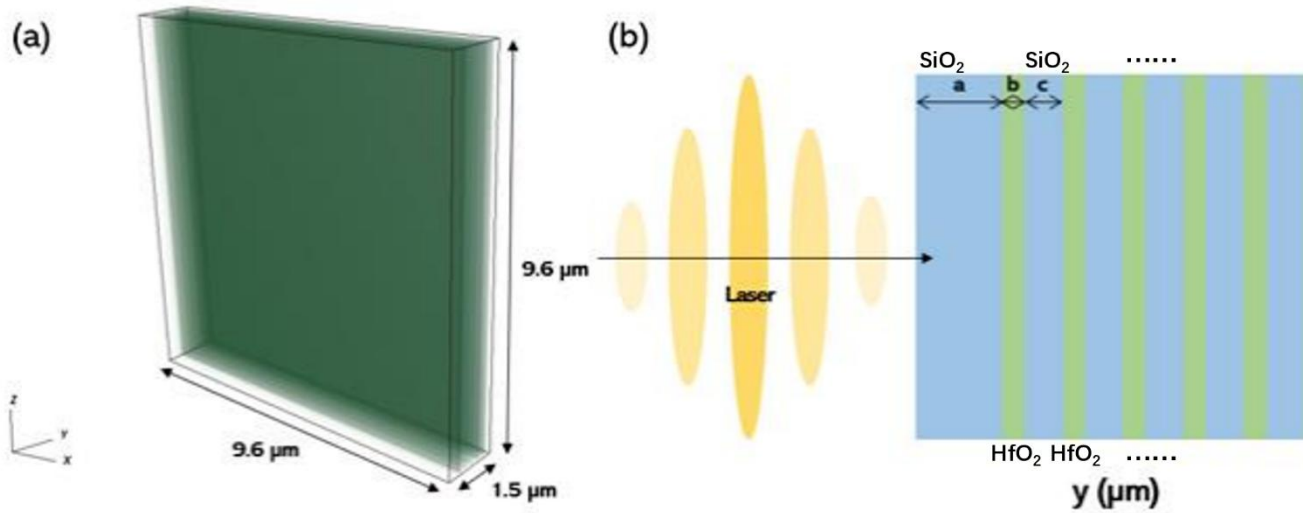
Simulation Set-up

Source
Wavelength: 800 nm
Duration: 7 fs
Waist: 4.65 μm
Gaussian Single Pulse, Normal Incident

Materials Properties [8]

	$m*/m_e$	W_0(eV)	n	Mass Density (g/cc)
a-SiO2	0.56	9	1.47654	2.20
a-HfO2	0.55	5.7	2.02122	9.68

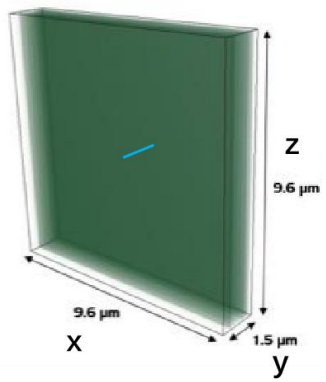
Resolution
40 nm (x, z): 15 nm (y)
1000 Macroparticle per cell
Loaded SiO_2 : 3.9 billion
Loaded HfO_2 : 2.1 billion



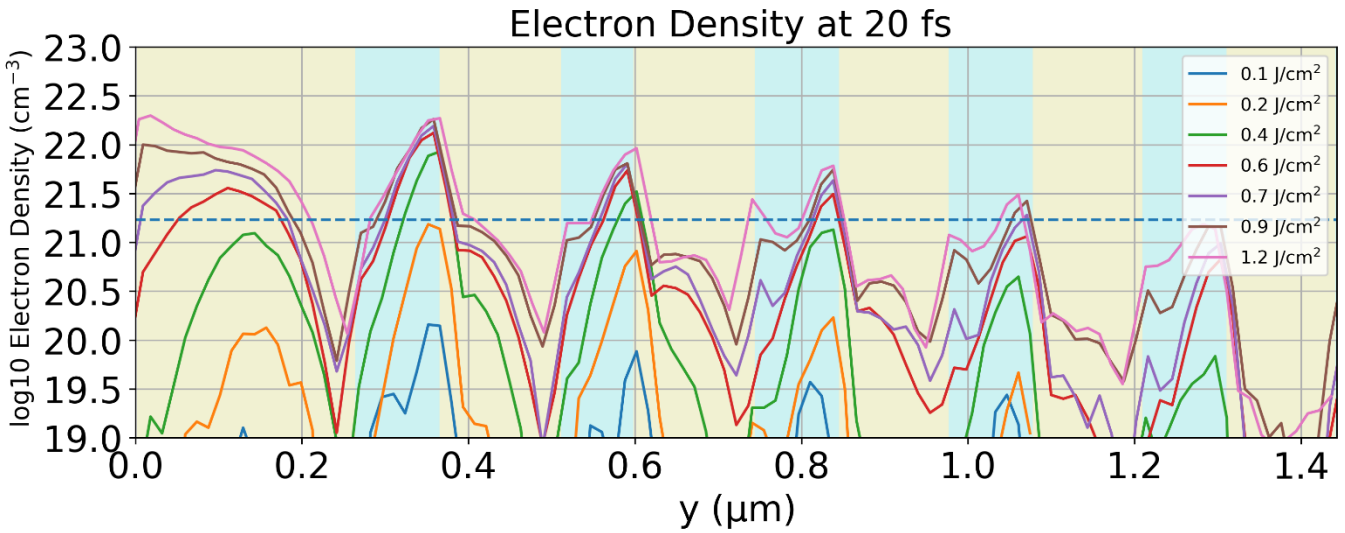
(a) 3D simulation window. (b) 2D schematic pulse and quarter-wave mirror with a silica protective layer

[8] Zhang, S.; Tripepi, M.; AlShafey, A.; Talisa, N.; Nguyen, H.T.; Reagan, B.A.; Sistrunk, E.; Gibson, D.J.; Alessi, D.A.; Chowdhury, E.A. Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings. Opt. Express 2021, 29, 39983–39999.

LIDT Criteria

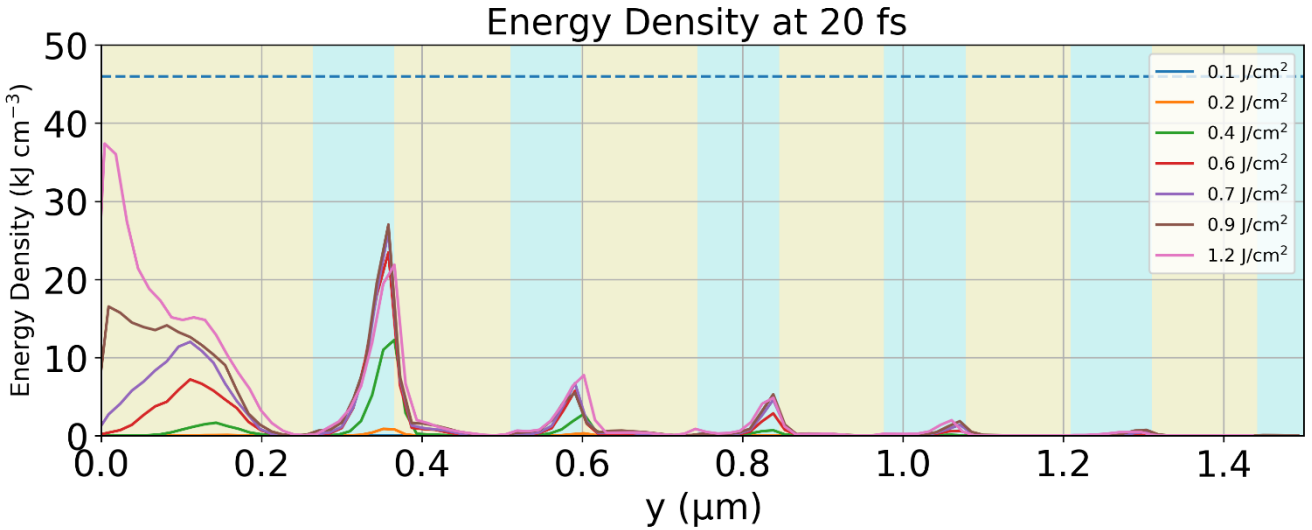


Critical Density



1.Critical Electron Density

The critical electron density at 800nm is about $1.7*10^{21}$ cm^{-3} , the damage is achieved at **0.2** J/cm^{-3}



2.Critical Energy Density

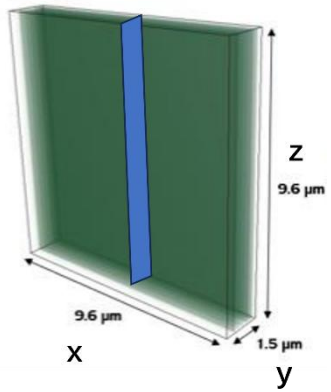
The dissociation energy of fused silica is 46-68 kJ/cm^3 [9,10]

[9] J. Peñano, P. Sprangle, B. Hafizi, W. Manheimer, and A. Zigler, "Transmission of intense femtosecond laser pulses into dielectrics," Phys. Rev. E 72(3), 036412 (2005).

[10] Kuan-Han Sun. Fundamental condition of glass formation. Journal of the American Ceramic Society, 30(9):277–281, 1947.

Electron Density Dynamics

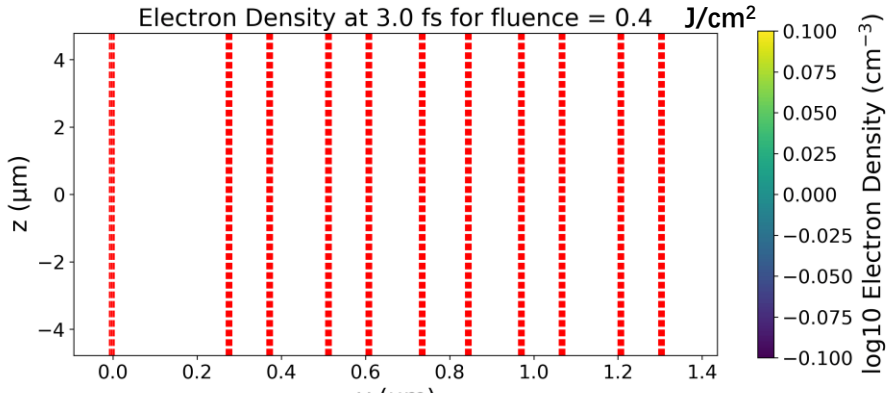
Electron density on y-z plane at the center of x.

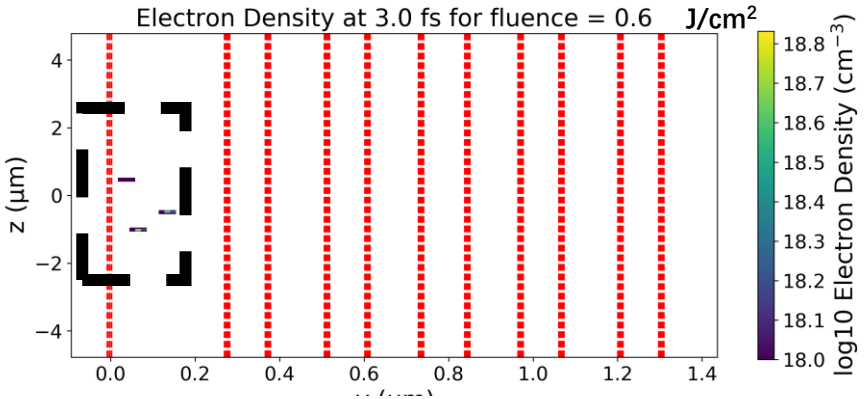


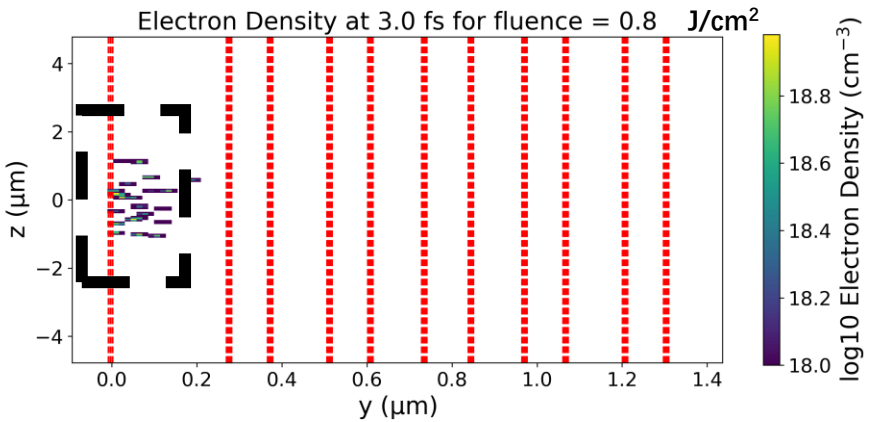
There is an expansion of the local peak electron density to the surface at higher fluence.

Implies

Plasma Generation







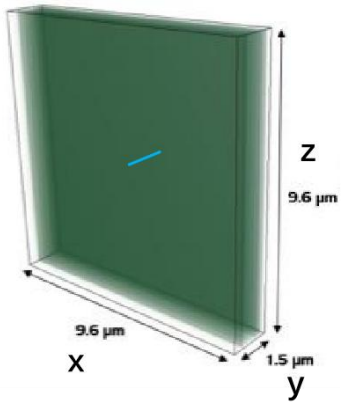
Electron Density Dynamics

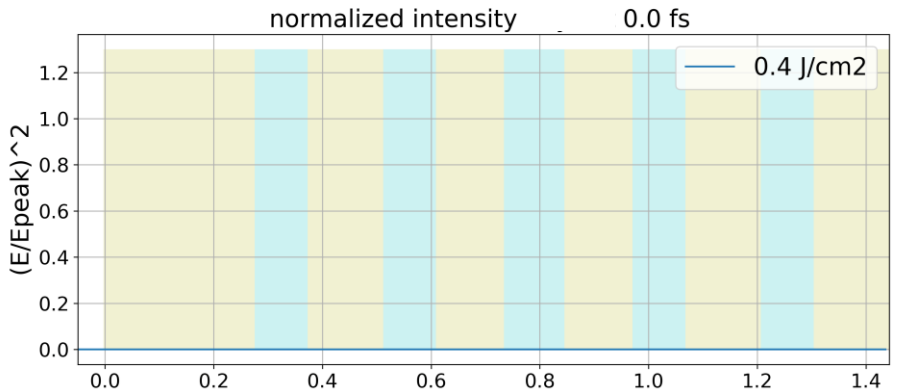
Plasma generation at the center of the top SiO_2 layer

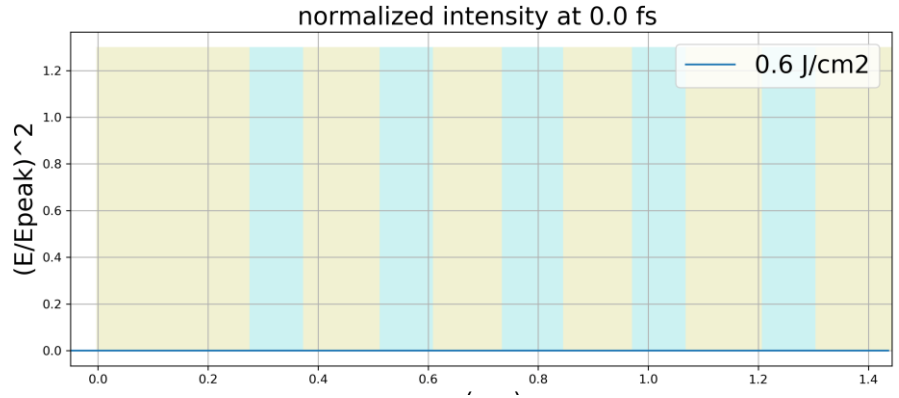
The plasma alters the optical properties and increases the refractive index

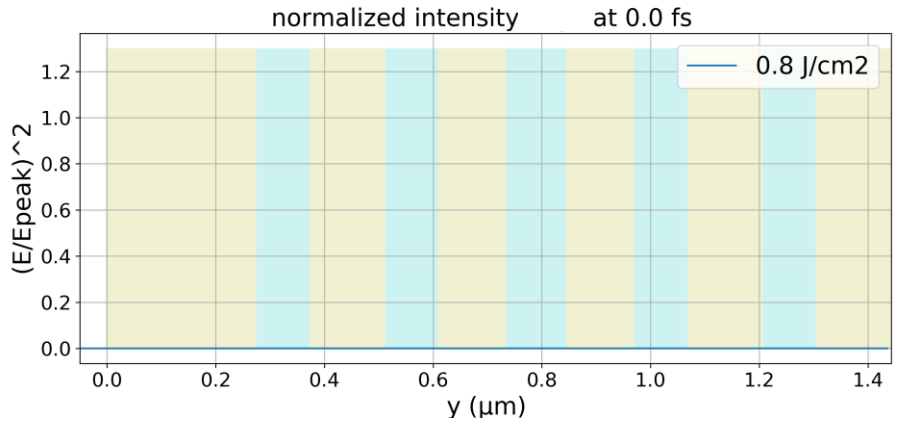
The resonant electric field pattern is changed and thus shifts the constructive peak to the surface

The ionization is then enhanced due to the high intensity





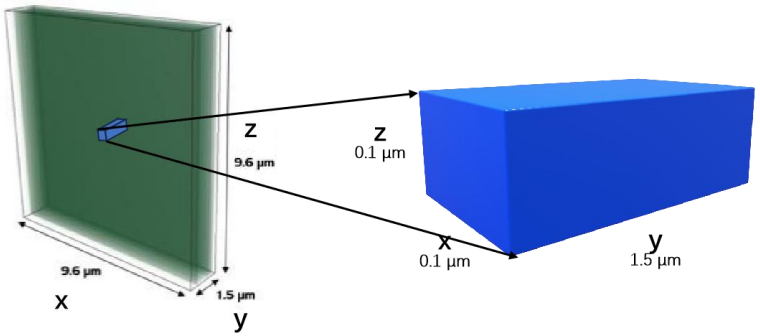




Electron Thermalization

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Fluence of 0.8 J/cm^2

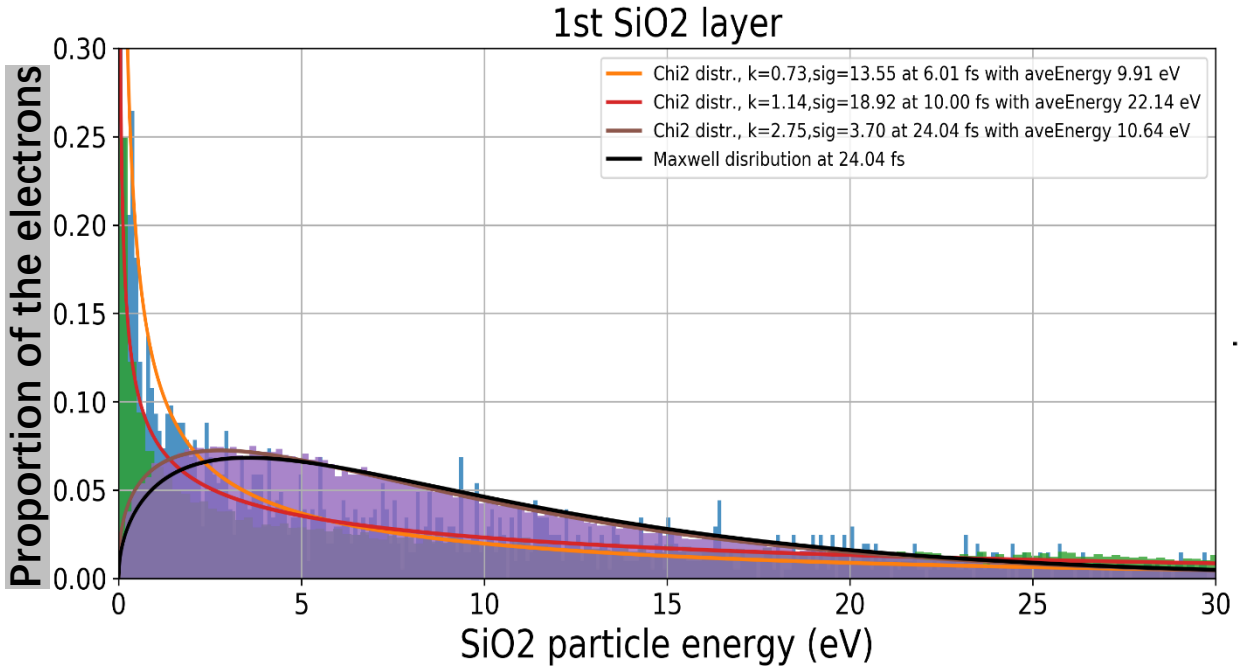


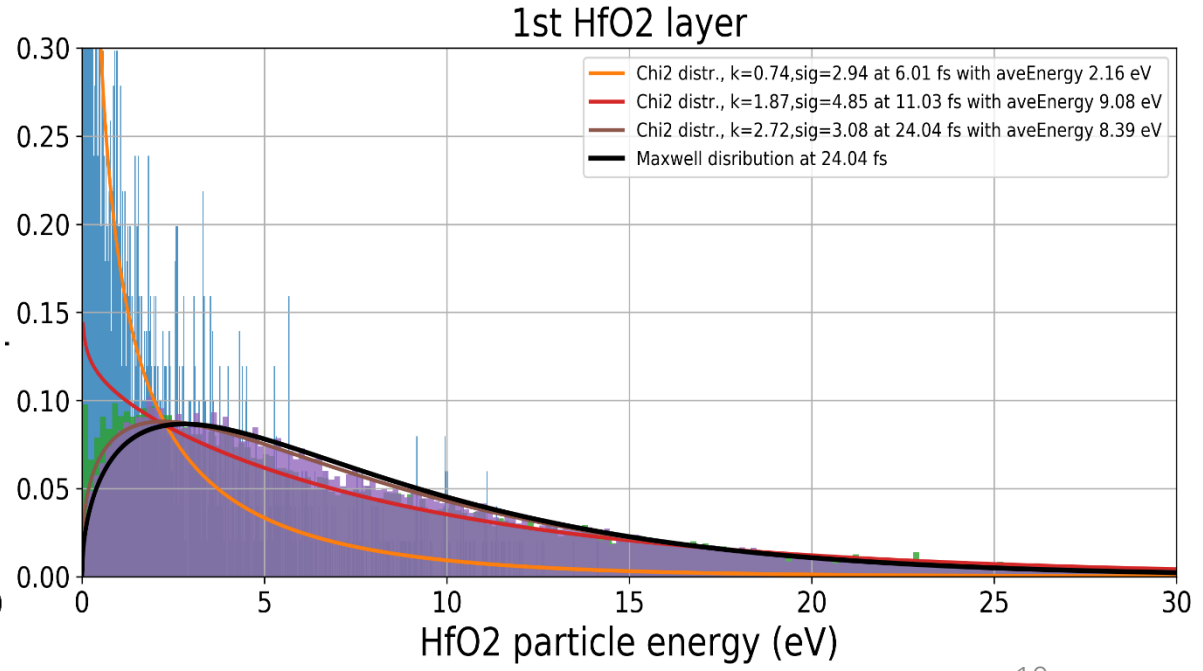
$$f(x;k,0,\sigma)=\frac{1}{\sigma^2}\frac{x^{\left(\frac{k}{2}-1\right)}e^{-\frac{x}{2\sigma^2}}}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)}$$

$$x=mv^2/2$$

$$\sigma\sim\sqrt{K_b*T}$$

When k = 3, the Chi^2 distribution becomes the Maxwellian distribution

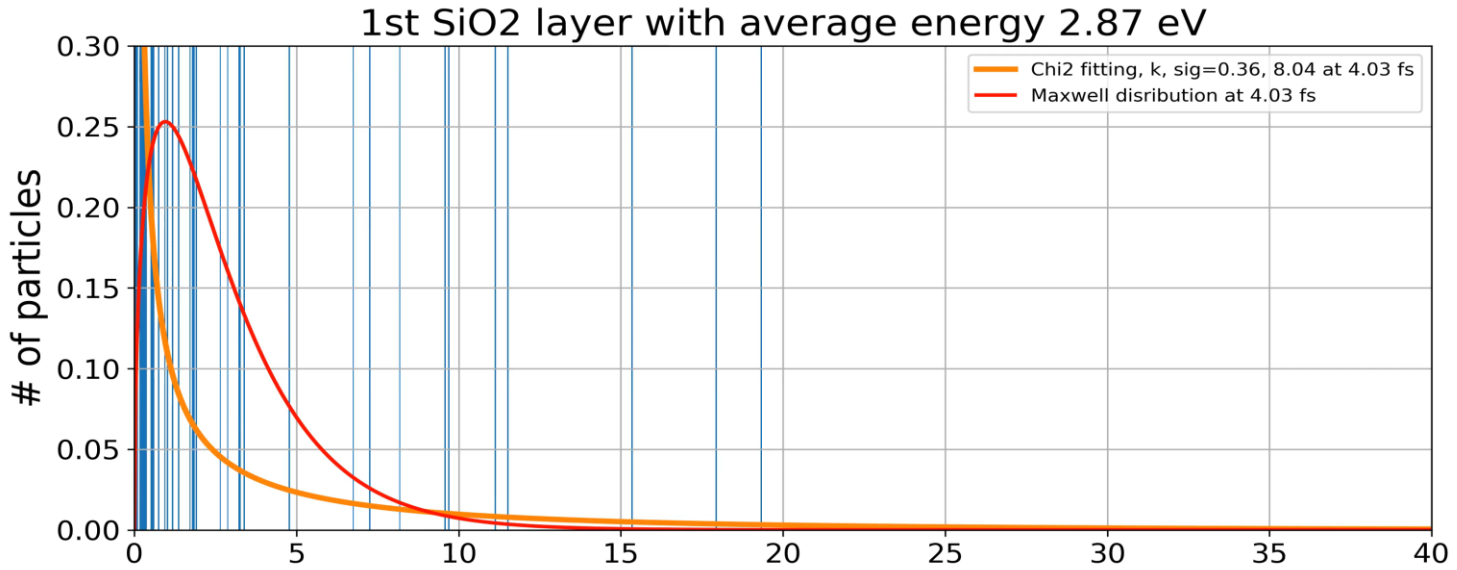


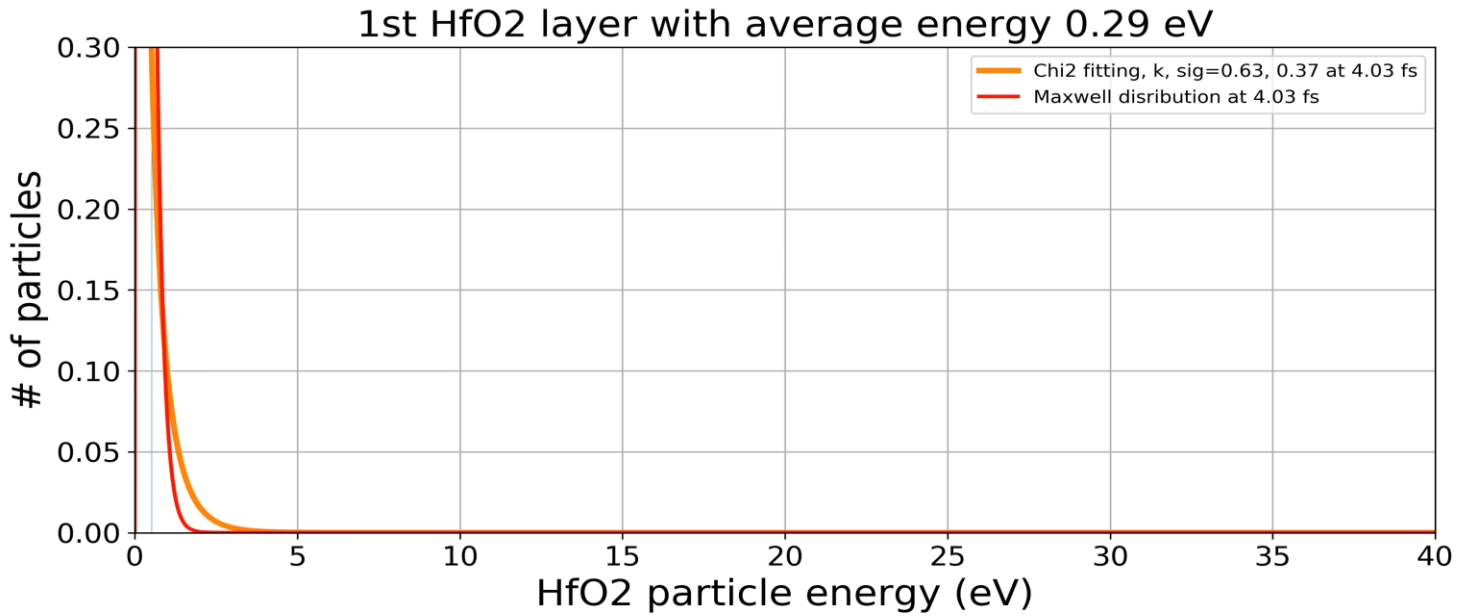


Electron Thermalization

Thermalization of the conduction band electrons is completed within about 15 fs.

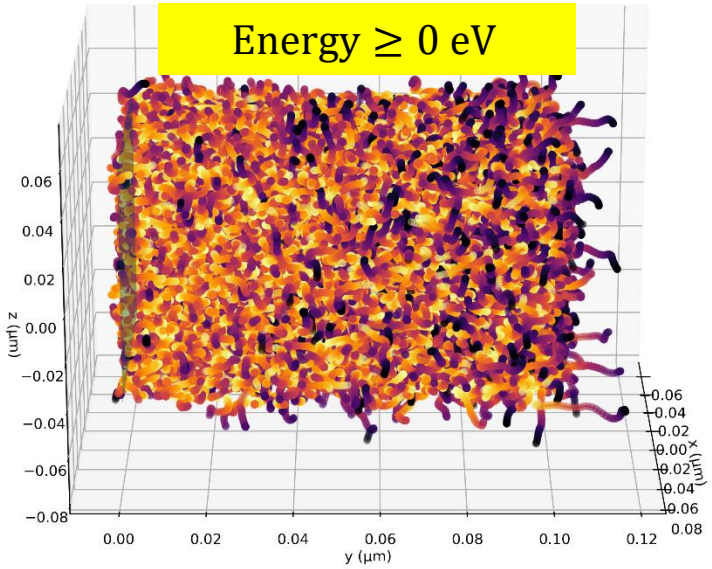
The oscillation of the distribution and average energy is because the total number of ionized electrons is increasing.

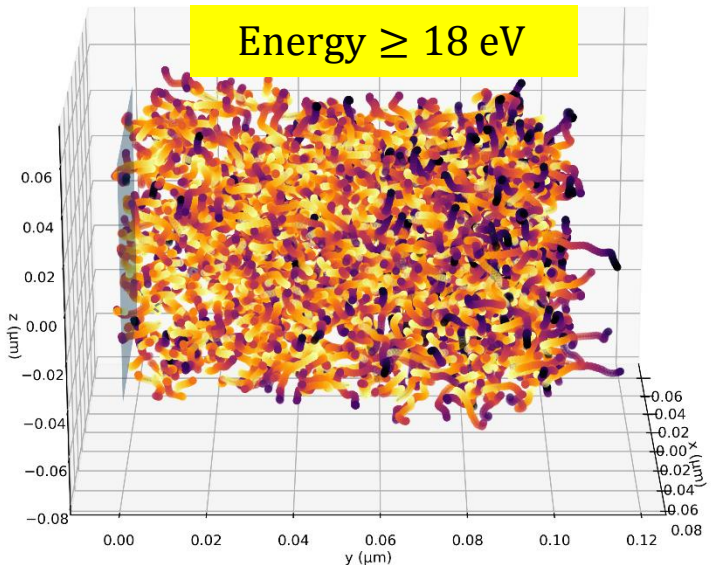


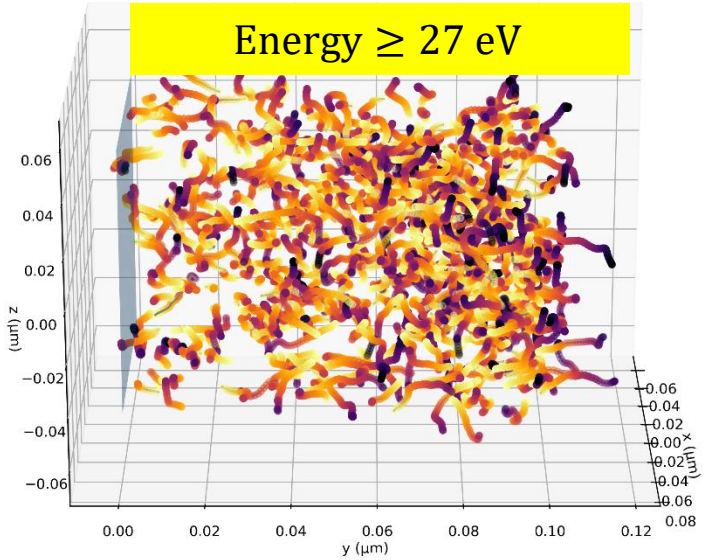


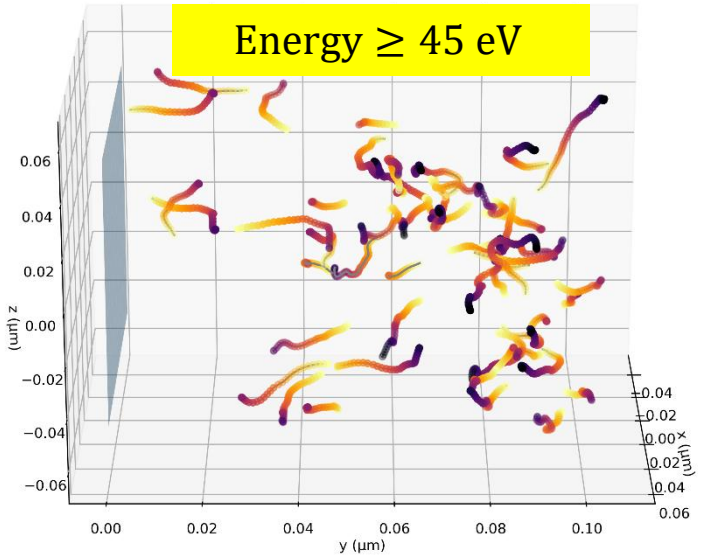
Electron Displacement

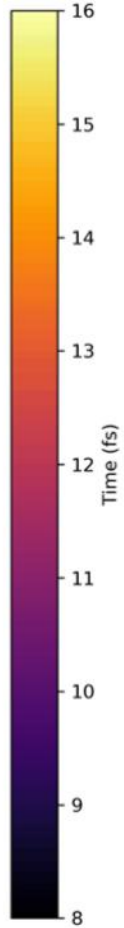
The track of conduction band electrons near air-SiO_2 interface from 8 fs to 16 fs



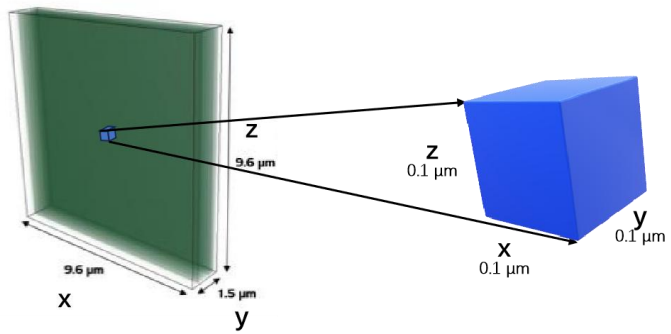






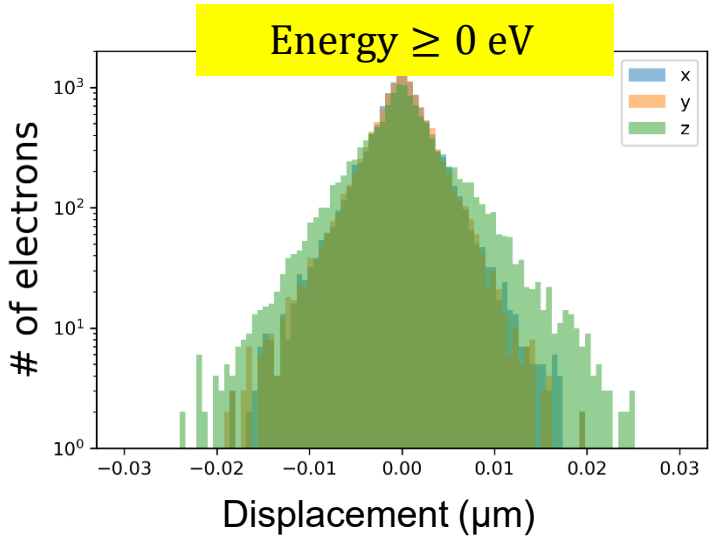


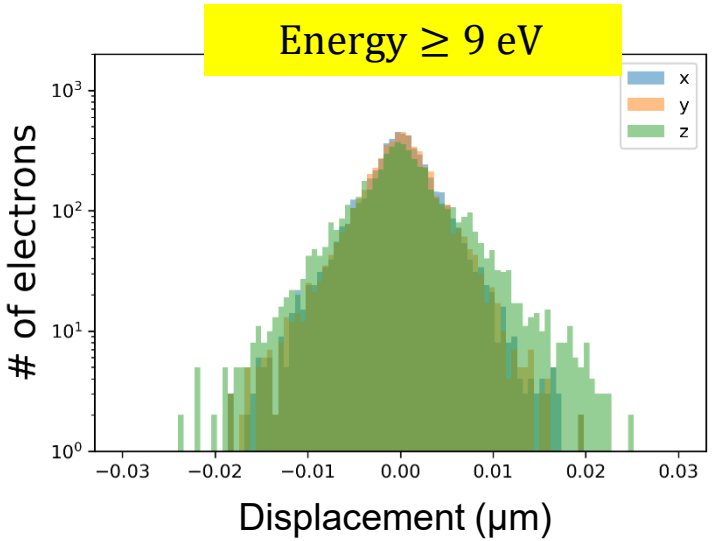
Fluence of 0.8 J/cm^2
Ionization energy for fused silica is 9 eV

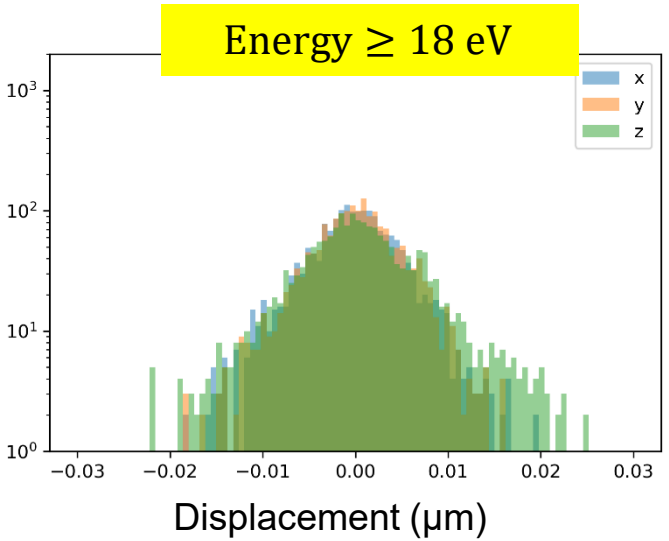


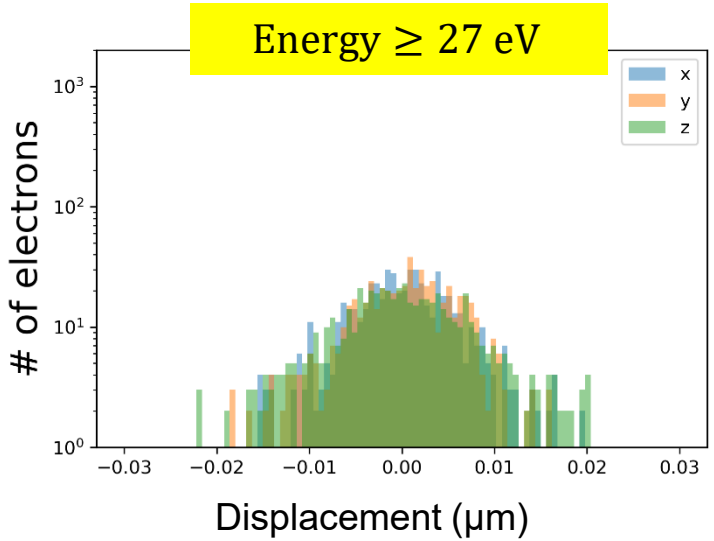
The conduction band electrons near the air-SiO_2 interface are born more lately, while the electrons near the center of top SiO_2 layer are born earlier

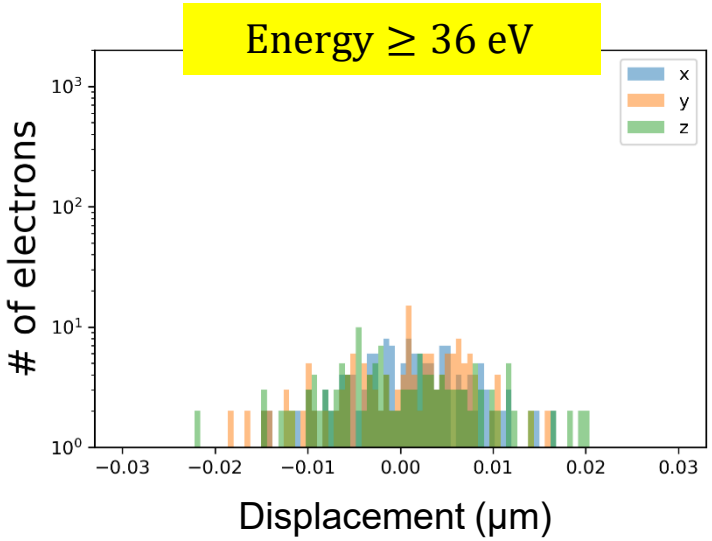
Electron Displacement

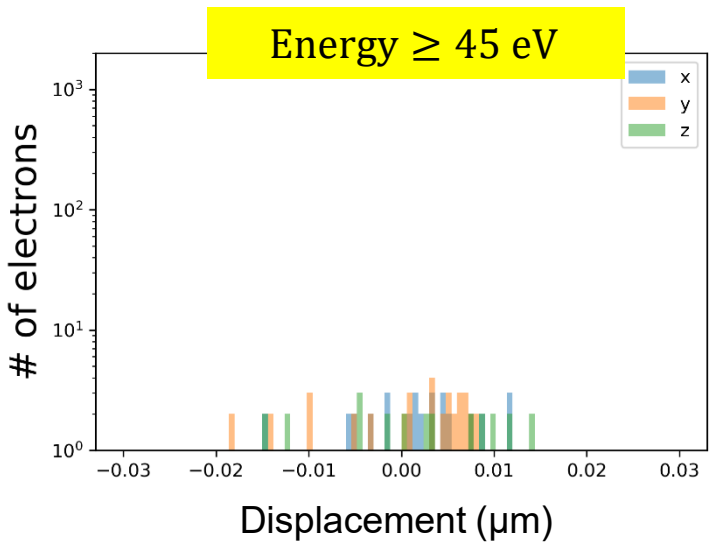












The polarization is along z

Summary:

1. Implemented Keldysh photoionization model and spatially varying refractive index in EPOCH
2. Different criteria give different conclusion
3. Implied the plasma generation
4. Revealed thermalization and polarization of the electrons

Outlook:

1. Impact Ionization Model

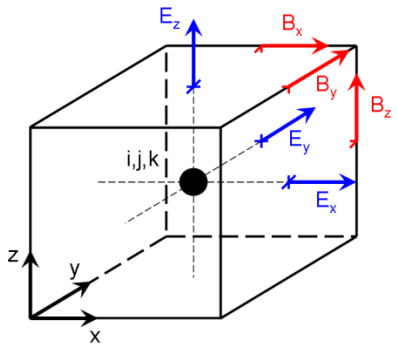
Thank you

Bibliography

[1] Matthias F. Kling, REPORT OF THE BASIC RESEARCH NEEDS WORKSHOP ON LASER TECHNOLOGY, 2023
[2] Kane Yee, "Numerical solution of initial boundary value problems involving maxwell's equations in isotropic media," in *IEEE Transactions on Antennas and Propagation*, vol. 14, no. 3, pp. 302-307, May 1966, doi: 10.1109/TAP.1966.1138693.
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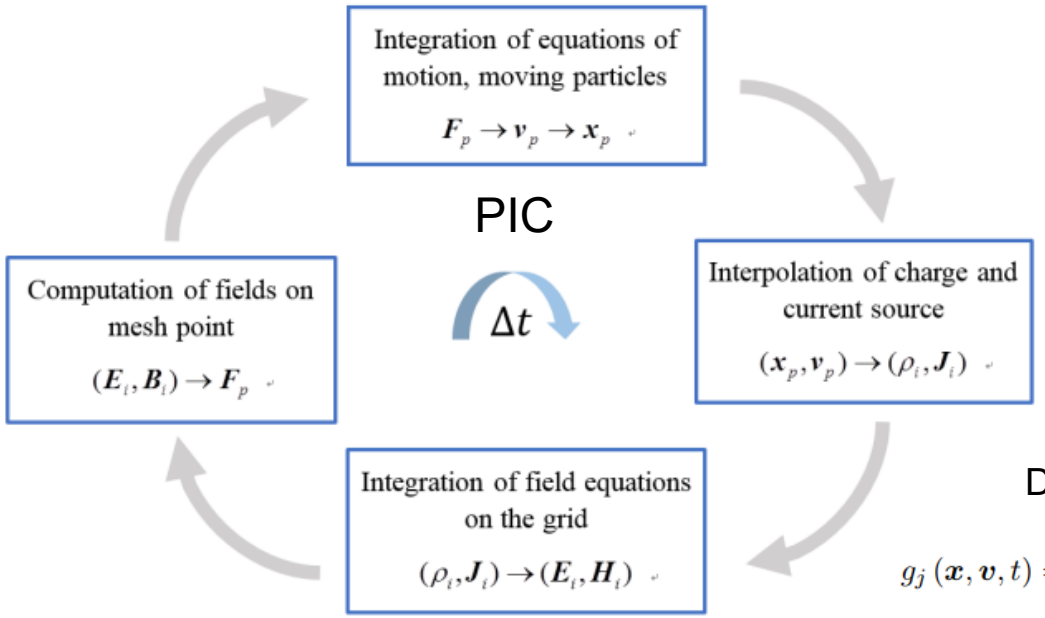
Back-up

Method Particle In Cell (PIC)



$$\begin{cases} \dfrac{d\boldsymbol{r}}{dt} = \boldsymbol{v} \\ \dfrac{d\boldsymbol{v}}{dt} = \dfrac{q}{\gamma m}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \end{cases}$$

Particle solvers



$$\nabla \cdot \boldsymbol{E} = \frac{\rho(\boldsymbol{x}, t)}{\epsilon_0},$$

$$\nabla \cdot \boldsymbol{B} = 0,$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t},$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J}(\boldsymbol{x}, t) + \frac{1}{c^2}\frac{\partial \boldsymbol{E}}{\partial t}.$$

Maxwell equations

Distribution function

$$g_j(\boldsymbol{x}, \boldsymbol{v}, t) = \sum_{i=1}^{N_j} [\delta(\boldsymbol{x} - \boldsymbol{r}_i(t))\, \delta(\boldsymbol{v} - \boldsymbol{V}_i(t))]$$

$$\rho(\boldsymbol{x}, t) = \sum_j \left(q_j \int g_j(\boldsymbol{x}, \boldsymbol{v}, t)\, \mathrm{d}\boldsymbol{v} \right)$$

$$\boldsymbol{J}(\boldsymbol{x}, t) = \sum_j \left(q_j \int \boldsymbol{v} g_j(\boldsymbol{x}, \boldsymbol{v}, t)\, \mathrm{d}\boldsymbol{v} \right)$$